

Sean Conway

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<https://sp-conway.github.io/sp-conway/>

Research Interests

Decision-making; choice; categorization; mathematical models.

Positions

2025-present Post-doctoral researcher, Purdue University, West Lafayette, IN (Stephen Broomell, advisor)

Education

2025 Ph.D., University of Massachusetts Amherst, Amherst, MA (Cognitive Psychology, Andrew Cohen, advisor)
 2019 M.S., University of Wisconsin Oshkosh, Oshkosh, WI (Psychology, Justyna Olszewska, advisor)
 2017 B.A., University of Wisconsin-La Crosse, La Crosse, WI (Psychology, minor in Music)

Publications

Journal Articles

- Conway, S.P.**, & Cohen, A.L. (2025). Testing principles of choice in categorization. *Computational Brain & Behavior*, 8, 442-469. <https://doi.org/10.1007/s42113-024-00232-6>
- Hiris, E., **Conway, S.P.**, McLoughlin, W., & Yang, G. (2021). Individual observer differences in the use of form and motion to perceive the actor's sex in biological motion displays. *Perceptual and Motor Skills*, 129(1), 5-32. <https://doi.org/10.1177/00315125211052923>
- Olszewska, J., Falkowski, A., **Conway, S.P.**, Sidoruk-Blach. M., & Jablonska, M. (2021). Ideal and anti-ideal as framing in political evaluations. Psychological analysis of political candidates in the theory of similarity. *Current Psychology*, 42(10), 8313-8326. <https://doi.org/10.1007/s12144-021-02168-7>

Conference Articles

- Abdulghani, Y., Wagner, C., Broomell, S.B., **Conway, S.P.**, & Guthrie, E. JWA-PyTorch: an Open-Source PyTorch-Based Toolkit for the Joint Weighted Average Operator. In *2026 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. IEEE.

Articles Submitted for Publication

Conway, S.P., & Cohen, A.L. (under review, invited revision). Correlated values as a mechanism for the repulsion effect.

Articles In Prep.

Conway, S.P., & Broomell, S.B. (invited revision in prep.). The case for more theoretically precise treatment of stochasticity in deterministic theories of choice and categorization.

Conway, S.P., & Broomell, S.B. (in prep.). Disentangling stochastic responding from perceptual discrimination in categorization.

Conway, S.P., & Cohen, A.L. (in prep). Multiple stimulus context effects in categorization choices.

Guthrie, E., Abdulghani, Y., **Conway, S.P.** Wagner, C., & Broomell, S.B. (in prep). Modeling the dynamics of human judgment with evidence weighted aggregation.

Conway, B.J., Taquet, L., Boerger, T.F., **Conway, S.P.**, Goetschel, K., Young, S.C., Krucoff, K.B., Schmidt, B.D., & Krucoff, M.O. (in prep). Differentiating patients with brain tumors from healthy adults using finger individuation.

Other Publications

National Postdoctoral Association. (2026). *2026 Institutional Policy Report*.

https://cdn.ymaws.com/www.nationalpostdoc.org/resource/resmgr/docs_v2/2026_NPA_Institutional_Polic.pdf [Data analysis and figure design for finished report].

Presented Talks

Conway, S.P., & Broomell, S.B. (2026, Nov.). Disentangling stochastic responding from perceptual discrimination in categorization. Talk accepted for presentation at the 67th *Annual Meeting of the Psychonomic Society*. San Diego, CA.

Conway, S.P., & Broomell, S.B. (2026, July). The Generalized Context Model, Luce's Choice Rule, and response stochasticity. Talk presented at the *2026 Meeting of the Society for Mathematical Psychology*. Montreal, CA.

Conway, S.P., Cohen, A.L., & Starns, J.J. (2025, June). Correlated valuations as a mechanism for the repulsion effect. Talk presented at the *2025 Virtual Meeting of the Society for Mathematical Psychology*.

Conway, S.P., & Cohen, A.L. (2023, June). Category biases arise from learning sets but not from choice sets. Talk presented at the *2023 Virtual Meeting of the Society for Mathematical Psychology*.

Conway, S.P., & Cohen, A.L. (2022, July). Categorization with multiple items: Empirical and modeling results. Talk presented at the *2022 Virtual Meeting of the Society for Mathematical Psychology*.

Poster Presentations

Conway, S.P., & Cohen, A.L. (2025, Nov). Violations of independence in best-worst choice. Poster presented at the *66th Annual Meeting of the Psychonomic Society*. Denver, CO.

Conway, S.P., Cohen, A.L., & Starns, J.J. (2025, July). Correlated valuations as a mechanism for the repulsion effect. Poster presented at the *58th Annual Meeting of the Society for Mathematical Psychology*. Columbus, OH.

Conway, S.P., Cohen, A.L., & Starns, J.J. (2024, Nov). Capturing value and covariance in choice, with an application to context effects. Poster presented at the *65th Annual Meeting of the Psychonomic Society*. New York, NY. [Poster selected for Psychonomic Society Graduate Travel Award]

Conway, S.P., & Cohen, A.L. (2023, Nov). Category biases arise from learning sets but not from choice sets. Poster presented at the *64th Annual Meeting of the Psychonomic Society*. San Francisco, CA.

Conway, S.P., & Cohen, A.L. (2023, Nov). Value covariance and the attraction-repulsion continuum. Poster presented at the *2023 Meeting of the Society for Judgment and Decision Making*. San Francisco, CA.

Conway, S.P., & Cohen, A.L. (2022, Nov). Categorization with multiple items: Empirical and modeling results. Poster presented at the *63rd Annual Meeting of the Psychonomic Society*. Boston, MA.

Conway, S.P., & Cohen, A.L. (2021, Nov). Testing the effect of choice set on categorization. Poster presented virtually at the *62nd Annual Meeting of the Psychonomic Society*.

Olszewska, J., **Conway, S.P.**, Falkowski, A., Jablonska, M., & Sidoruk-Blach, M. (2019, Nov). Ideal and anti-ideal as framing in political evaluations. Psychological analysis of politicians' evaluations in the light of the contrast model of similarity. Poster presented at the *60th Annual Meeting of the Psychonomic Society*, Montreal, Canada.

Conway, S.P., Mohawk, K., Timmins, R., & Lishner, D. (2019, Apr). An examination of replication variability across seven replication studies. Poster presented at the *Ninety-First Annual Meeting of the Midwestern Psychological Association*, Chicago, IL.

Olszewska, J., **Conway, S.P.**, Falkowski, A., Jablonska, M., Barnes, J., Hodel, A., & Michels, S. (2019, Apr). Do comparisons matter? Valence framing and evaluation of political

candidates. Poster presented at the *Ninety-First Annual Meeting of the Midwestern Psychological Association*, Chicago, IL.

Hiris, E., McLoughin, W., Yang, G., & **Conway, S.P.** (2018, May). Individual differences in the use of form and motion in the perception of sex in biological motion displays. Poster presented at the *2018 Meeting of the Vision Sciences Society*, St. Pete Beach, FL.

Invited Talks

Conway, S.P. (2025, Dec.). *The Generalized Context Model, Luce's Choice Rule, and response stochasticity*. Talk presented at The University of Illinois Urbana-Champaign Quantitative Psychology Colloquium. Champaign, IL.

Conway, S.P. (2025, Feb.). *Generating the repulsion effect through correlated valuations*. Talk presented at The University of Missouri Quantitative Psychology Colloquium. Columbia, MO.

Conway, S.P. (2023, Oct). *Context effects in decision-making*. Talk presented at University of Massachusetts Amherst First Year Seminar on Decision-Making. Amherst, MA.

Awards

Psychonomic Society Graduate Travel Award

- 65th Annual Meeting of the Psychonomic Society, Nov. 2024

Summer School on Modeling Heterogeneity: Advances in Behavioral Decision Analytics

- Summer 2023 – Awarded spot in two-week workshop modeling behavioral choice data. Innsbruck, Austria. PIs Michel Regenwetter and Daniel Cavagnaro.

University of Massachusetts Amherst – Data Analytics and Computational Social Science Program – Graduate Non-Working Fellowship

- Fall 2021, Fall 2022

University of Massachusetts-Amherst - University Graduate Fellowship

- Spring 2020

Teaching History

University of Massachusetts Amherst – Instructor

- Methods of Inquiry (Research Methods) in Psychology (Summer 2025)
- Statistics in Psychology (Summer 2024)
- Data Science Fundamentals (Winter 2022, Summer 2022, Winter 2023, Summer 2023, Winter 2024, Winter 2025)
- Junior Year Writing (Spring 2022)

University of Massachusetts Amherst – Lab Instructor

- Statistics in Psychology (Fall 2020, Spring 2021, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025)

Ad Hoc Journal Reviews

- *Journal of Experimental Psychology: Learning, Memory, and Cognition*
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Other Activities

National Postdoctoral Association – Volunteer data analysis and visualization for 2025
Institutional Policy Report

- March 2026 – July 2026

University of Wisconsin Oshkosh Graduate Student Association – Public Relations Officer

- Fall 2018 – Spring 2019 Academic Year

University of Wisconsin Oshkosh Graduate Council – Graduate Student Representative

- September 2018 – May 2019
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